

Underground Economy and Corruption



Course Project: CS752
Instructor: **Prof. Om Damani**

Presented by

Piyush Sawarkar (22M0772)

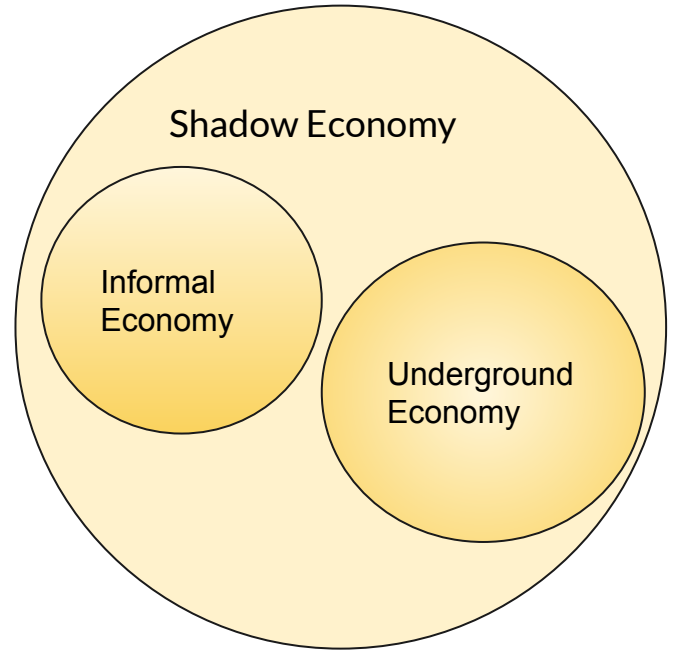
Sayan Surya Shaw (22M0760)

Chiranmoy Bhattacharya (22M0744)

Introduction:

The Shadow Economy consists of all economic activities that are hidden from official authorities for monetary, regulatory, and institutional reasons, that is, all activities that contribute to the official gross national product (GNP) but are not reported to tax authorities.

Shadow Economy represents the effect of both Underground economy and Informal Economy.

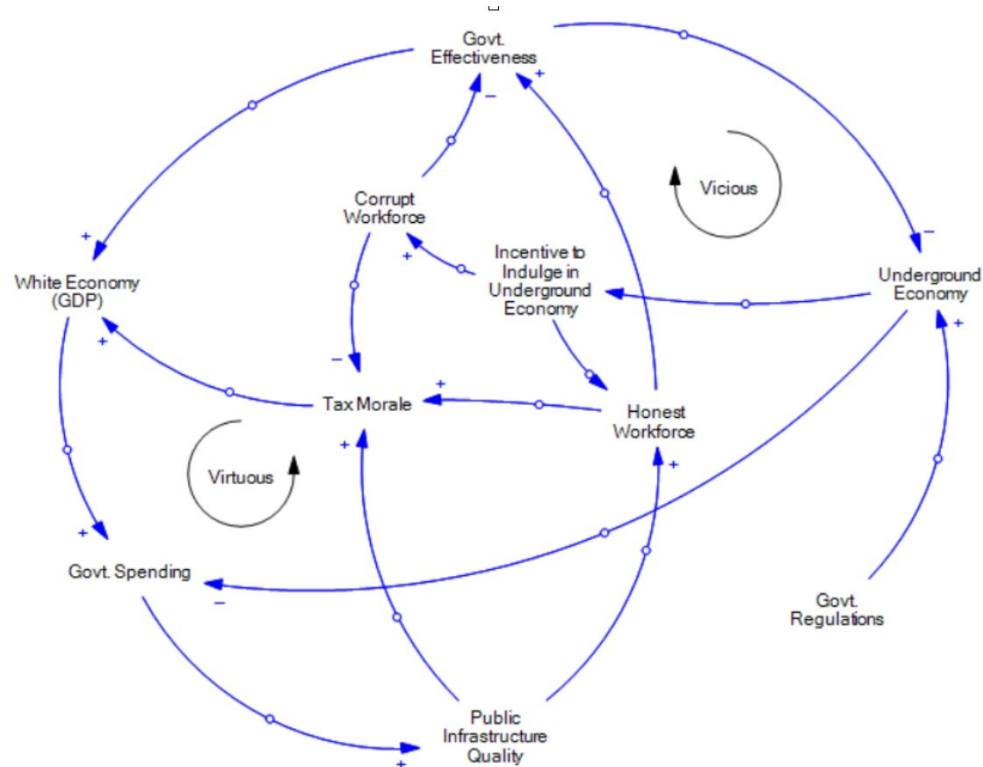




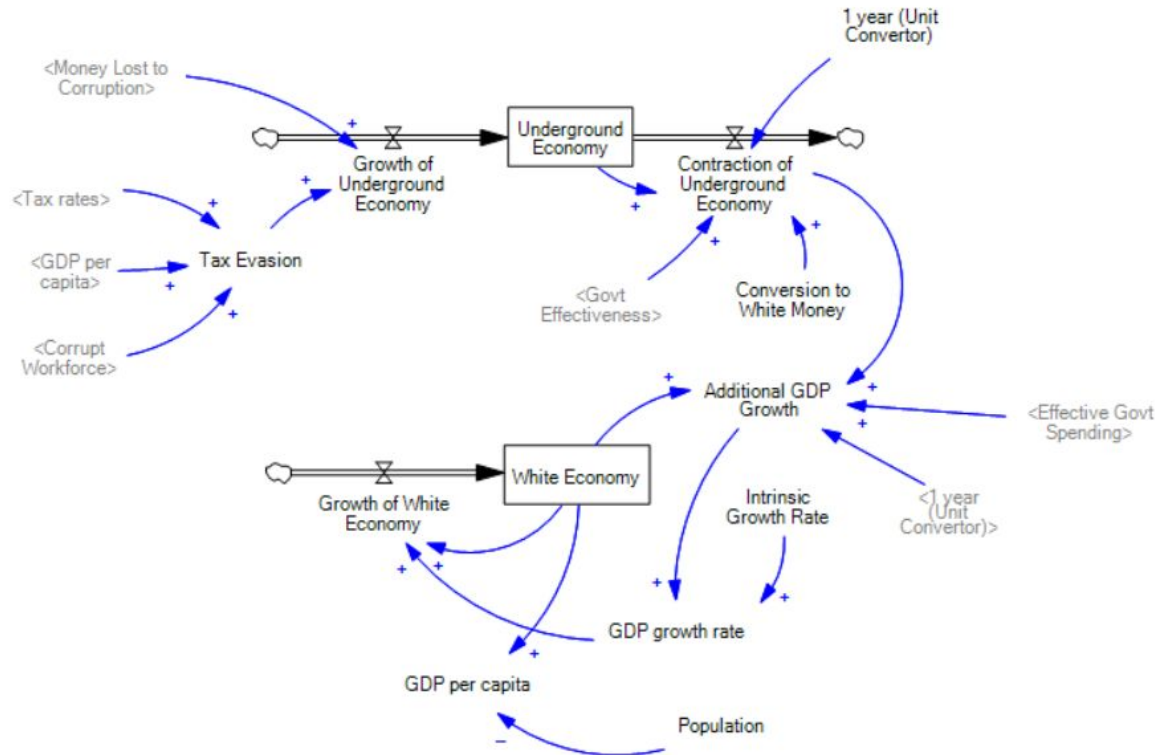
Motivation:

- Underground Economy is a very important topic when the question of economic growth of a country comes to picture.
 - Even though it is discussed a lot, its exact structure and behaviour is not known to the common public.
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- General trend is to blame underground economy for everything suboptimal in our society.
 - To understand how actually underground economy works in deep we are modelling this topic.

Causal Loop Diagram:



Underground Economy and White Economy

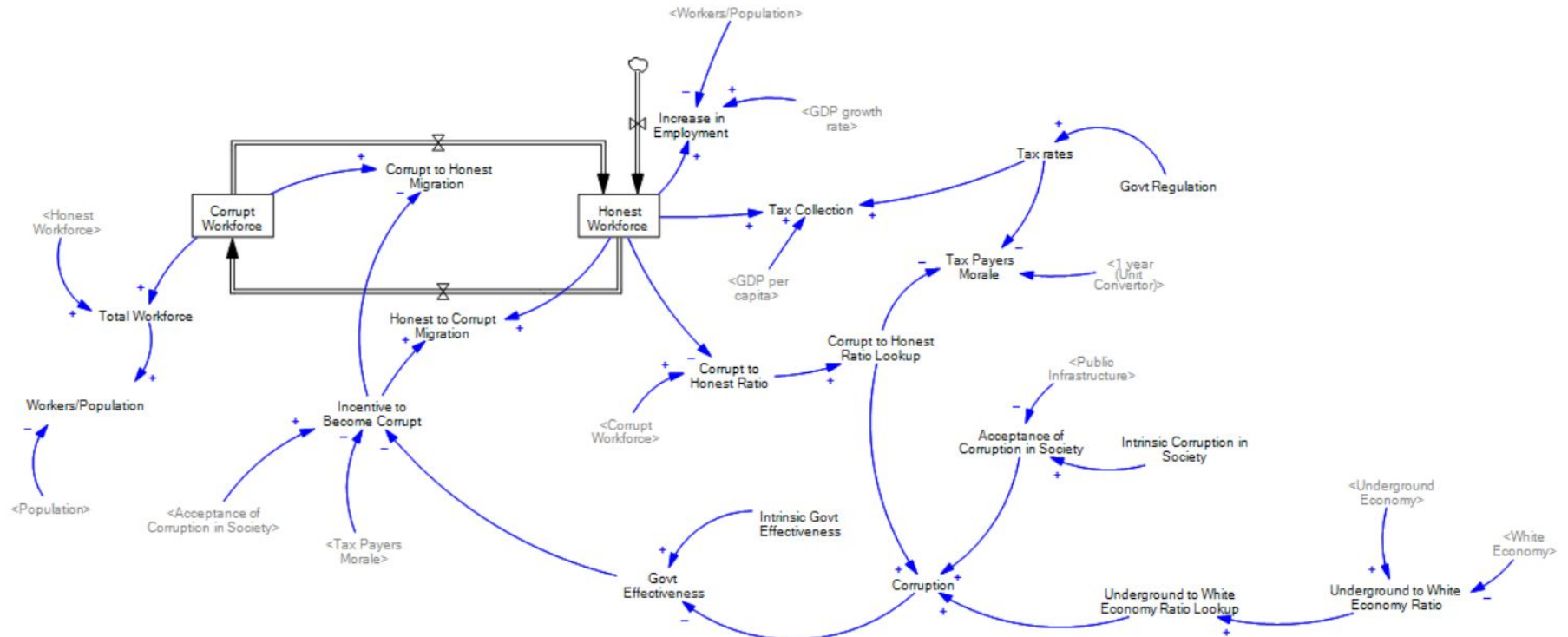


Major Equations & Units:



1. Contraction of Underground Economy = $\text{Underground Economy} \times (\text{Conversion to White Money} + \text{Govt Effectiveness} / 100) \times 1 \text{ year (Unit Converter)}$
Units: Rupee/Year
2. GDP per Capita = $\text{White Economy} / \text{Population}$
Unit: Rupee/Year
3. GDP Growth Rate = $(\text{Additional GDP Growth} / 5 + \text{Intrinsic Growth Rate}) / 100$
Unit: dmnl/Year
4. Additional GDP Growth = $(\text{Effective Govt Spending} + \text{Contraction of Underground Economy}) / \text{White Economy} \times 1 \text{ year (Unit Converter)}$
Unit: dmnl/Year

Corrupt and Honest Workforce



Major Equations & Units:



1. $\text{Corruption} = (\text{Acceptance of Corruption in Society} + 0.7 * \text{Corrupt to Honest Ratio Lookup} + 1.3 * \text{Underground to White Economy Ratio Lookup}) / 3$

Units: Dmnl

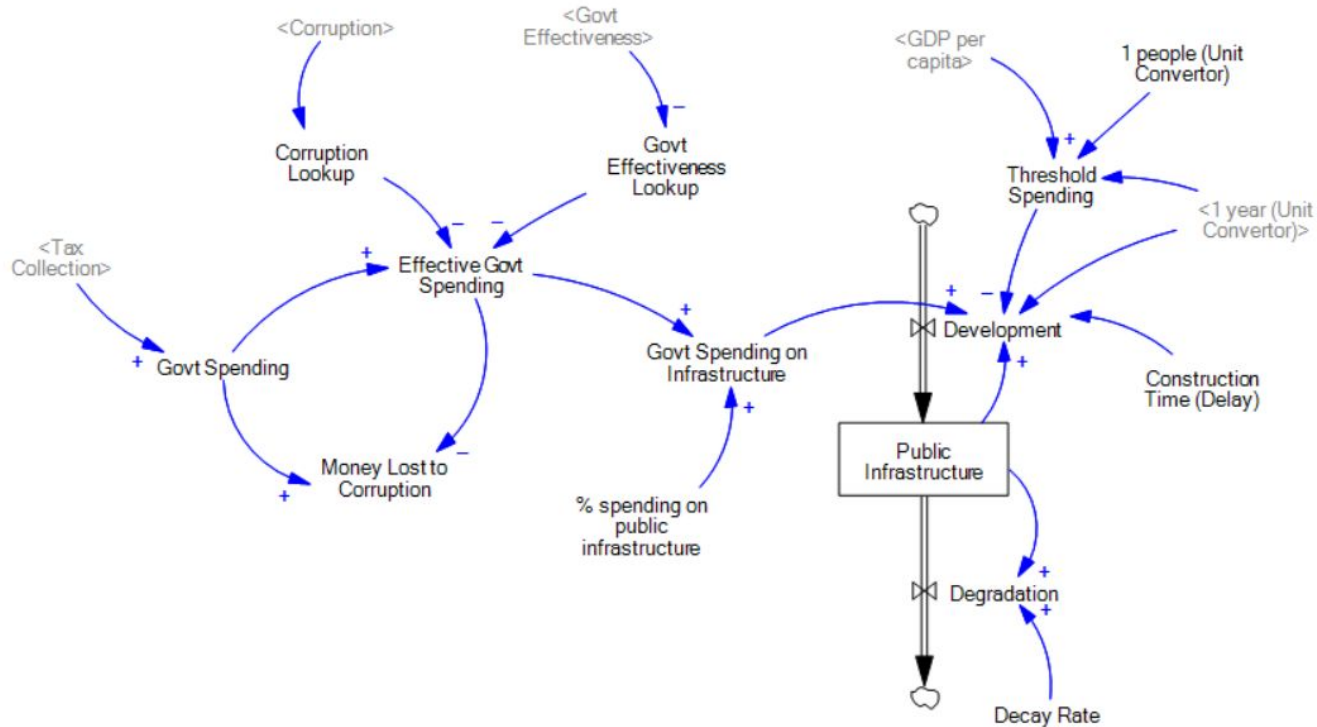
2. $\text{Incentive to Become Corrupt} = \text{WITH LOOKUP} ((4 * \text{Acceptance of Corruption in Society} + (10 - \text{Tax Payers Morale}) * 2 - 2 * \text{Govt Effectiveness}) / 7,$

Units: Dmnl/Year

3. $\text{Tax Collection} = \text{Honest Workforce} * \text{Tax rates} * \text{GDP per capita} / 100$

Units: Rupee/Year

Public Infrastructure

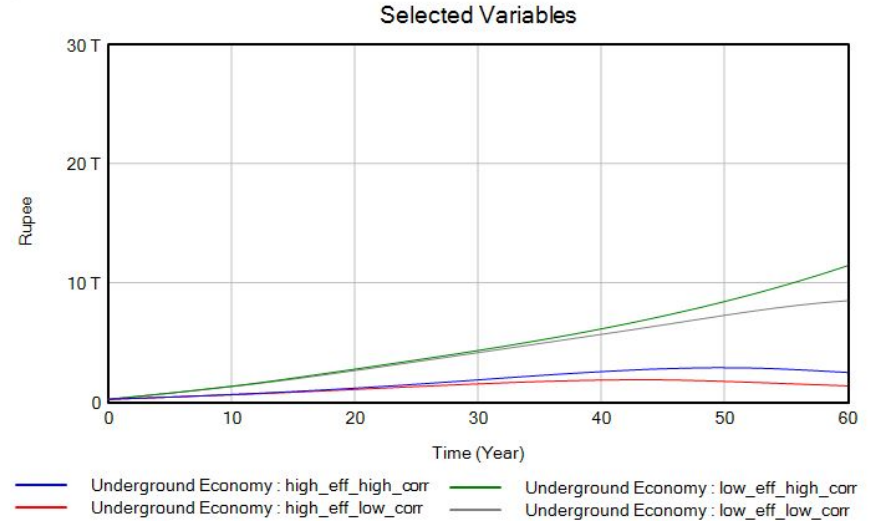
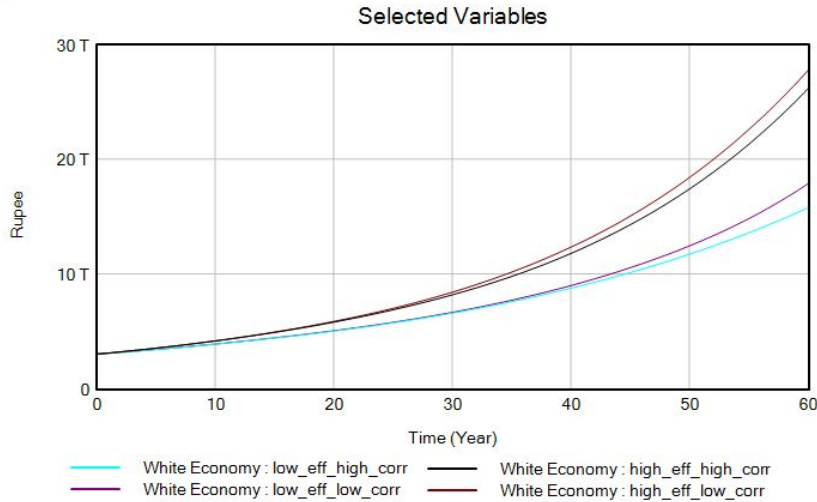


Major Equations & Units:

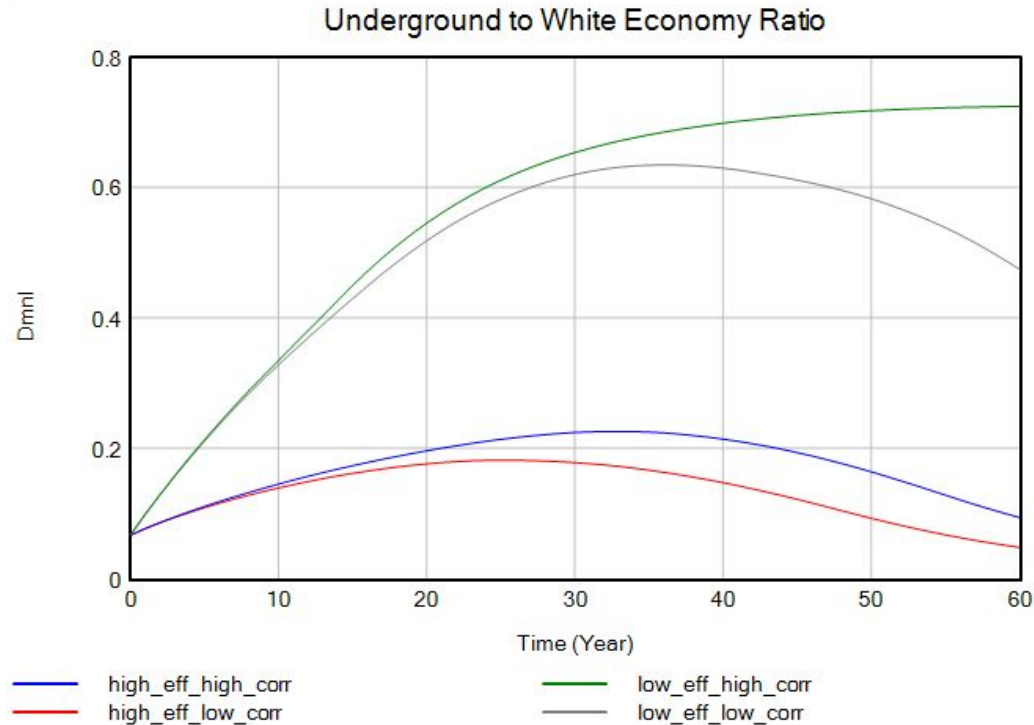


1. $\text{Development} = \text{DELAY3}(\text{Public Infrastructure} * \tanh(\text{Govt Spending on Infrastructure} / \text{Threshold Spending}) / 10, \text{"Construction Time (Delay)"}) / 1 \text{ year (Unit Converter)}$
Units: Dmnl/Year
2. $\text{Threshold Spending} = (\text{GDP per capita} * \alpha) * 1 \text{ people (Unit Converter)} / 1 \text{ year (Unit Converter)}$
Units: Rupee/Year , $\alpha = 10^4 * 7$
3. $\text{Effective Govt Spending} = \beta * \text{Govt Spending} * (10 - (\text{Corruption Lookup} + \text{Govt Effectiveness Lookup}) / 2)$
Units: Rupee/Year, $\beta = 0.1$

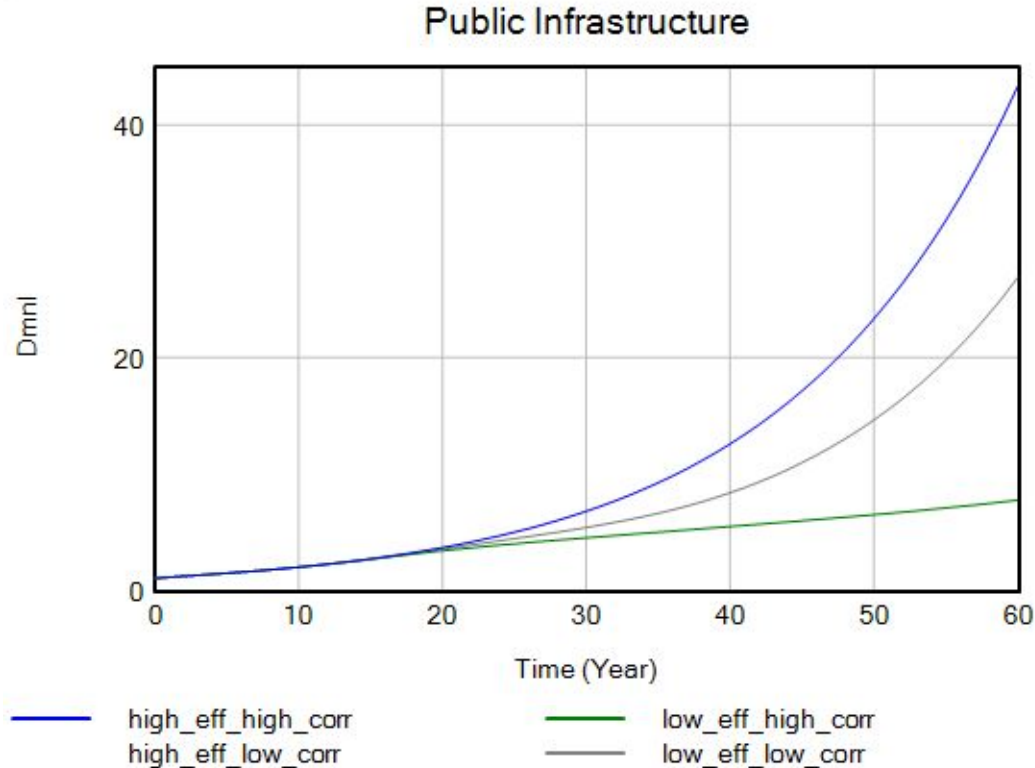
Effects of Govt. Efficiency and Corruption on Economy



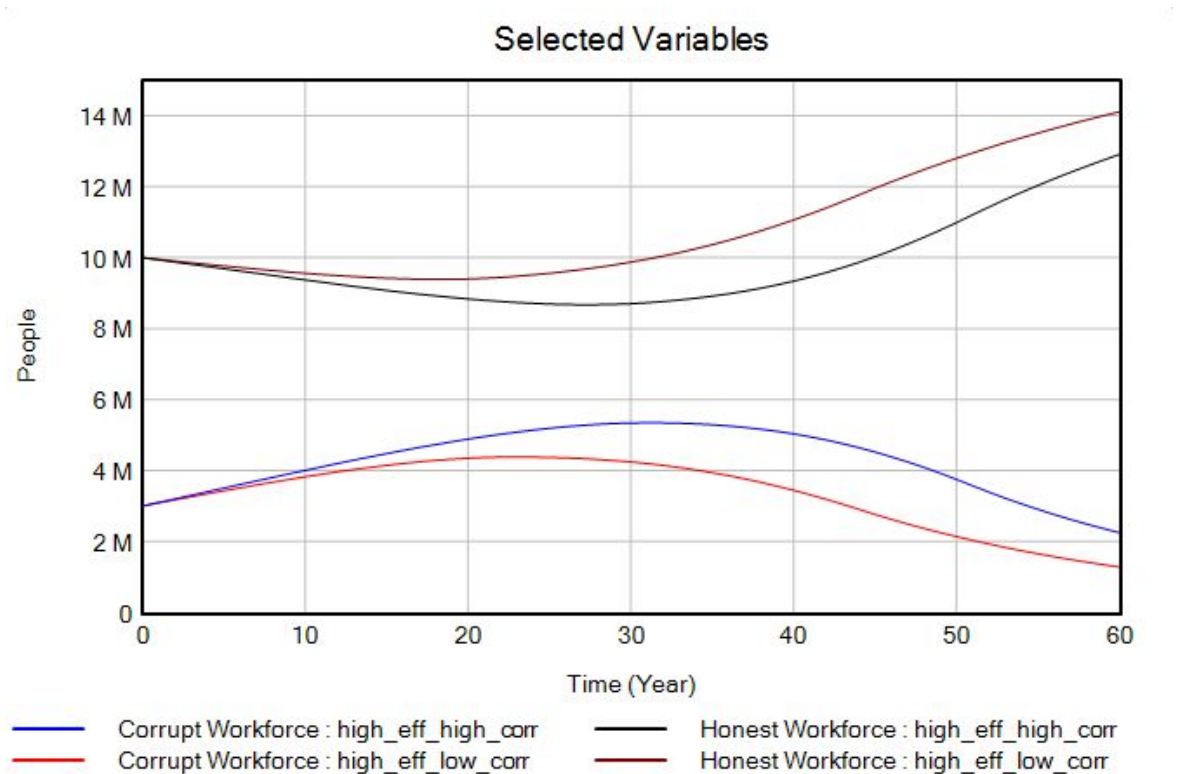
Underground to White Economy Ratio



Effects of Govt. Efficiency and Corruption on Infrastructure



Effects of Govt. Efficiency and Corruption on Workforce





Data Source:

- For this model we have considered the annual data of 51 countries from 2000- 2005.
- All data is publicly available and is provided by international organizations such as the World Bank.
- Among the 51 countries included in the sample, 10 are OECD countries and 6 are non-OECD “high income countries” as defined by the World Bank. The remaining countries are emerging markets [1].
- The countries can be grouped into two categories: *industrialized countries* and *developing or emerging countries*.



Problems faced:

- Identifying Key Variables and Grouping Similar Variables.
- Determining the Impact of Variables on the Growth or Decline of a Stock.
- Identifying the Relation between Cause & effects.
- Setting Up the equations with hyper-parameters
- Modelling Soft Variables.



Conclusions from the Model:

- The underground economy is seen to increase due to high corruption and lower law and regulation against bribery and illegal factors.
- Conversion of black money to white money is also considered in many scenarios, leading to formal or legal money being counted as part of the underground economy.
- From the modeling graphs, we can conclude that effective government rule is more important than corruption for controlling underground economy.
- In the future, we will try to apply the model to real national datasets and consider additional factors specific to each country to predict results accurately.



Reference:

1. Dell' Anno, Roberto and Friedrich Schneider (2003): The shadow economy of Italy and other OECD countries: What do we know?, Journal of Public Finance and Public Choice, 21/2-3, pp. 97-120.
2. Shadow Economies and Corruption All Over the World: What Do We Really Know?
Link: <https://www.iza.org/publications/dp/2315/shadow-economies-and-corruption-all-over-the-world-what-do-we-really-know>
3. The Shadow Economy: Friedrich Schneider & Colin C. Williams, The Shadow Economy, Institute of Economic Affairs, 2013 Link: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2286334
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Thank You